Cheng Chen

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Education

Institute of Automation, Chinese Academy of Sciences, Beijing, China — Sept. 2027 (Expected)
Incoming Ph.D. Student; Supervisor: **Prof. Kaiqi Huang**

Tongji University, Shanghai, China — Sept. 2023 –Jun. 2027
B.Eng. in Communication Engineering (expected Jun. 2027)
Major Rank: **3/85**

Publications

- If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization
  ACM MM 2026 (Accepted).
- OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment
  IJCAI 2026 (Accepted).
- Neural-Operator Reliability and Risk Triage
  NeurIPS 2026 (Under Review).

Research Experience

If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization

Accepted at **ACM MM 2026**, Co-first Author

- Addressed modality reliability asymmetry in open-vocabulary audio-visual event localization, where audio is semantically useful but often off-screen, delayed, or noisy, so symmetric distillation lets weak-audio supervision distort temporal boundaries.
- Proposed **OV-OrthKD**, a reliability-aware asymmetric distillation framework: a strong visual teacher supervises decision-aligned features, we restrict a weak audio teacher to an auxiliary orthogonal subspace, and a text prototype anchors open-vocabulary transfer.
- On OV-AVEBench, reached **0.816** segment AP and improved F1@0.5 by **+2.7** points overall and **+3.4** on unseen categories over the official fine-tuning baseline; ablations show weak-audio feature transfer helps only when kept off the decision path.

OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment

Accepted at **IJCAI 2026**, Co-first Author

- Addressed expertise asymmetry in multi-teacher distillation for diabetic retinopathy screening, where a weaker ViT teacher (**0.83 QWK**) harms students under standard logit fusion with a stronger CNN teacher (**0.88 QWK**).
- Proposed **OrthKD**, distilling only complementary feature-level context from the weak teacher while enforcing orthogonality between teacher projections to avoid redundant knowledge transfer.
- Evaluated on **132,049** retinal images; the **5.4M**-parameter MobileNetV3 student reached **0.885 QWK** on EyePACS and improved zero-shot Messidor-2 performance from **0.507** to **0.728 QWK** (**+22.1** percentage points).

Neural-Operator Reliability and Risk Triage

NeurIPS 2026, Under Review; First Author

- Studied deployment-time reliability for neural operators, where conventional uncertainty estimators require repeated inference and can erase the latency advantage of fast field prediction.
- Developed **Allocation Uncertainty Decomposition**, a single-pass risk score computed from prediction-path allocation entropy without posterior sampling or an auxiliary uncertainty head.
- On electromagnetic scattering, improved Spearman correlation from **0.61** to **0.67** and failure-detection AUROC from **0.78** to **0.82** at **1.13**× batched cost; selectively escalating high-risk cases reached **0.72** correlation and **0.85** AUROC.

Awards

- National First Prize, China Robot and Artificial Intelligence Competition (2026), Team Leader
- National Second Prize, Global Campus Artificial Intelligence Algorithm Elite Competition (2025), Team Leader
- National Third Prize, National Undergraduate Embedded Chip and System Design Contest (2025), Team Leader

Skills

Programming: Python, C/C++, Verilog.
Framework: PyTorch.
Tools: Git, Linux, LaTeX, OpenAI Codex, Claude Code.
Foundations: Machine Learning, Algorithms, Linear Algebra, Probability, Optimization, and Signal Processing.